

HEP: Hierarchical Ensemble Personalization for Efficient and Fair Federated Learning

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Abstract—Personalized Federated Learning (PFL) addresses statistical data heterogeneity (Non-IID data) across decentralized edge devices. However, practical deployments are governed by the Personalization Trilemma—the fundamental trade-off between statistical personalization accuracy across diverse skew regimes, on-device memory and compute constraints, and worst-case client fairness. Existing PFL paradigms make rigid compromises across this trilemma: Global Consensus methods degrade under severe statistical skew; Dual-Model Regularization achieves robust personalization but doubles local memory footprint and per-batch training latency; Decoupled/Split-Head methods remain parameter-efficient but suffer representation collapse under homogeneous (IID) data; and Clustered Topologies lack on-device local head specialization.

In this paper, we propose HEP (Hierarchical Ensemble Personalization), a parameter-efficient framework designed to navigate the multi-dimensional trade-offs of the Personalization Trilemma. HEP reconciles global consensus learning, cluster-level collaboration, and local edge specialization through a single shared-backbone 3-tier architecture (Root, Parent, Local). It dynamically balances training using an entropy-based label skew metric, binomial loss weighting, and active-class logit masking. To protect client privacy, HEP uses random projection sketching for data-free cluster routing, providing formal differential privacy guarantees and preventing gradient inversion attacks while preserving clustering geometry.

Across diverse heterogeneity regimes, HEP demonstrates a Pareto-optimal operating profile across the Personalization Trilemma: it achieves top-tier personalization accuracy under extreme statistical skew while matching global consensus models under homogeneous distributions, operates strictly within a single-backbone memory envelope at roughly half the memory footprint and latency of dual-model approaches, and significantly improves worst-case tail-client fairness. We additionally formalize the threat model boundary, demonstrate in-scope label fault containment without external heuristics, and report a negative result showing that server-side Byzantine filtering cannot substitute for architectural head isolation against gradient corruption on shared backbones. All design choices carry measured justifications, cross-backend reproducibility is verified, and all artifacts are publicly available.

Index Terms—Federated Learning, Edge Computing, Personalization Trilemma, Parameter-Efficient Learning, Hierarchical Ensembles, Differential Privacy.

I. Introduction

Federated Learning (FL) [1] enables distributed edge devices to collaboratively train statistical models without centralizing private raw data. Despite its privacy benefits, deploying federated optimization in production

edge systems reveals the Personalization Trilemma—a fundamental tension among three competing objectives:

- 1) **Statistical Generalization & Skew Resilience:** Maintaining high accuracy across diverse data distributions, ranging from extreme Non-IID label skew ($\alpha \leq 0.05$) to uniform homogeneous (IID) partitions.
- 2) **On-Device Hardware Constraints:** Operating within strict edge device budgets (< 150 MB Peak VRAM, minimal per-batch latency, and single-pass feature evaluation) without inducing out-of-memory faults.
- 3) **Worst-Case Fairness & Network Realism:** Ensuring that tail-decile (bottom 10%) clients do not suffer convergence failure under sparse sampling ($C_p \leq 0.20$) or intermittent network connectivity.

As summarized in Table I, existing Personalized Federated Learning (PFL) paradigms navigate this trilemma with rigid structural compromises:

- **Global Consensus FL (FedAvg, FedProx, SCAFFOLD):** Optimizes a single consensus model across all clients [1]–[3]. While hardware-efficient ($1\times$ model footprint), global models experience severe client drift and accuracy degradation when local data distributions diverge.
- **Dual-Model Regularization (Ditto, APFL):** Concurrently trains a global model and a private personalized model coupled via L2 proximal penalties [4] or interpolation scalars [5]. While achieving high personalization accuracy, maintaining two complete model graphs doubles client VRAM and per-batch compute passes, creating deployment barriers on memory-constrained edge hardware.
- **Decoupled / Split-Head Paradigms (FedRep, FedPer, FedBABU, FedALA):** Shares a global feature extractor and isolates private local classification heads [6]–[9]. While parameter-efficient, naive split-head models lack intermediate collaborative structure and experience substantial representation collapse under homogeneous (IID) data.
- **Clustered Topologies (Clustered FL):** Groups clients into sub-networks based on gradient similarity [10]. However, standard clustered FL enforces flat FedAvg within each cluster, lacking on-device local head specialization and failing to protect individual tail

Hardware Efficiency & Time-to-Accuracy Profile (CIFAR-10 ResNet9)

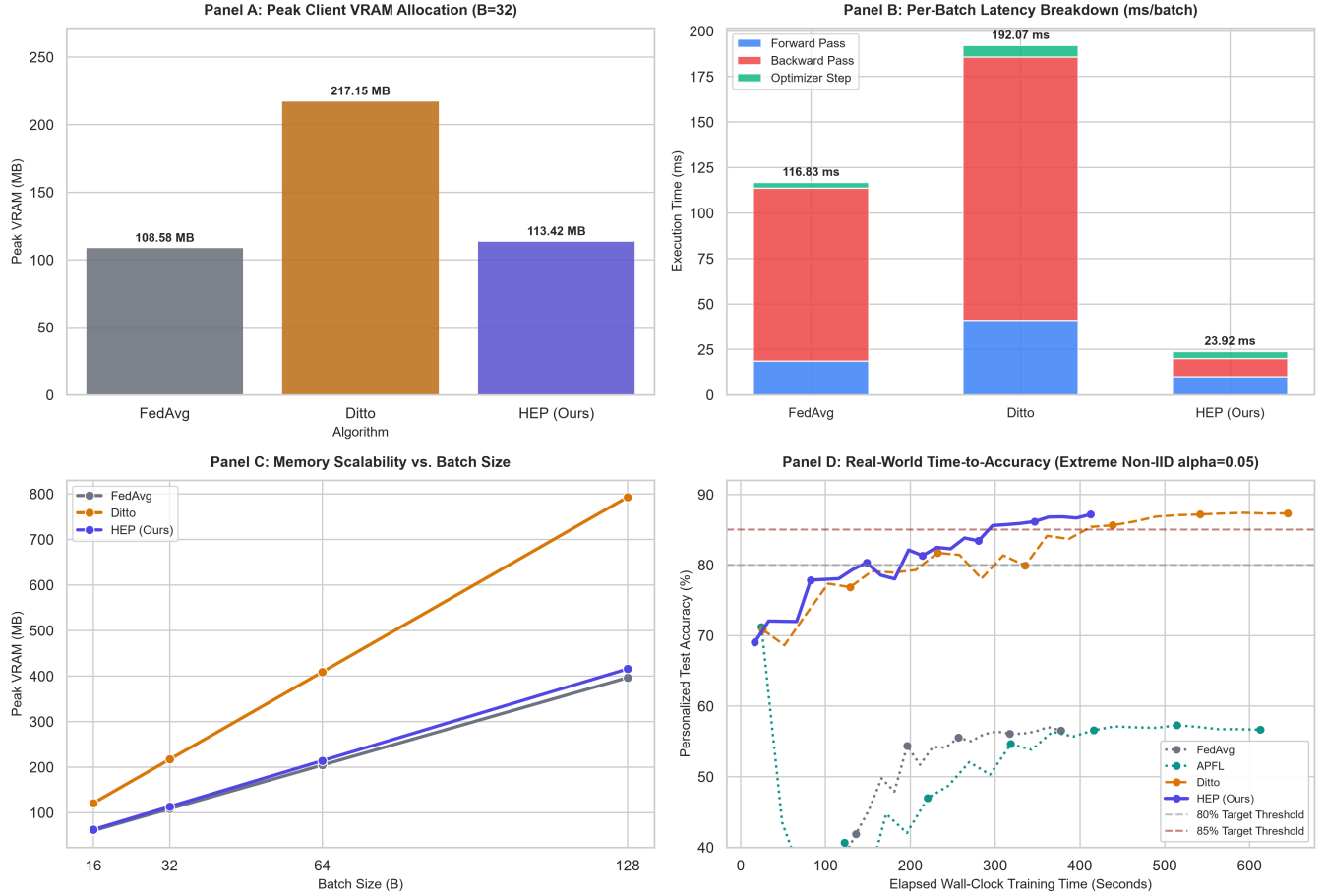


Fig. 1: Hardware Resource, Latency, and Wall-Clock Time-to-Accuracy Profile on CIFAR-10 ResNet9. Panel A: Peak client VRAM allocation at batch size $B = 32$, demonstrating a 47.8% memory reduction for HEP compared to dual-model Ditto (113.42 MB vs. 217.15 MB). Panel B: Per-batch millisecond latency breakdown across forward pass, backward pass, and optimizer update (HEP achieves an 8.35 ms/batch latency vs. Ditto’s 16.78 ms/batch, delivering a 50.2% speedup). Panel C: Memory footprint scalability across batch sizes ($B = 16 \rightarrow 128$), showing dual-model methods scaling at $2\times$ the memory slope of HEP. Panel D: Simulated wall-clock time-to-accuracy trajectory under extreme Non-IID skew ($\alpha = 0.05$), where HEP reaches 85% accuracy in 297s (116s faster than Ditto).

clients.

Rather than viewing PFL as a binary choice between resource-heavy dual models and fragile split-head architectures, we propose HEP (Hierarchical Ensemble Personalization), a lightweight framework that navigates the trade-offs of the Personalization Trilemma:

- **Shared-Backbone Multi-Head Multiplexing:** Edge clients maintain a single shared convolutional backbone paired with three lightweight classification heads: a global Root head, a cluster-level Parent head, and a private Local head. Feature representations are computed only once per batch.
- **Adaptive Loss Weighting & Routing:** Automatically balances training across the three heads via an entropy-based label skew metric ($R_{skew,i}$), binomial

loss weighting, variance-adaptive cluster momentum ($\beta_{c,k}$), and staleness-aware fallback routing.

- **Privacy-Preserving Cluster Routing:** Compresses linear head updates via 64-dimensional random projection sketching with local differential privacy, preventing gradient inversion attacks while preserving clustering geometry.
- **Intra-Cluster Semantic Sample Pooling:** Aggregates $6\times$ sample support (2000 pooled samples per class vs. 333 local samples under $\alpha = 0.05$), mitigating local overfitting while preserving tail fairness.

A. Summary of Key Contributions

- 1) **Multi-Paradigm Synthesis & Trilemma Formulation:** We formulate the Personalization Trilemma and characterize the structural trade-offs between dual-

TABLE I: Taxonomy and Structural Trade-offs across PFL Paradigms.

PFL Paradigm Family	Representative Methods	Memory Overhead	IID Robustness	Multi-Scale Hierarchy
Global Consensus	FedAvg, FedProx, SCAFFOLD	1× (Low)	High	None (Flat Global)
Dual-Model Regularization	Ditto, APFL	2× (High)	Moderate	None (Global + Private)
Decoupled / Split-Head	FedRep, FedPer, FedBABU, FedALA	1× (Low)	Poor (IID Collapse)	None (Body + Head)
Clustered Topologies	Clustered FL (CFL)	1× (Low)	Moderate	Sub-Network Only
Hierarchical Ensemble (HEP)	HEP (Ours)	1× (Low)	High (Entropy Calibrated)	3-Tier (Root, Parent, Local)

model personalization, split-head parameter efficiency, and clustered sub-networks.

- 2) Adaptive Hierarchical Personalization: We introduce HEP, deriving loss weights, staleness scales, cluster momentum, and temperature parameters dynamically, eliminating manual heuristic hyperparameter searches.
- 3) Privacy-Preserving Sketch Routing: We design a random projection sketching and local DP mechanism for cluster routing that prevents gradient inversion attacks while ensuring high routing stability.
- 4) Semantic Sample Pooling Analysis: We quantify how update-similarity clustering enables intra-cluster sample pooling under extreme skew ($6\times$ data support, 2000 pooled examples vs 333 local samples per class), preventing localized overfitting.
- 5) Hardware & Compute Efficiency: HEP operates within a single-backbone envelope, retaining 47.8% lower peak VRAM (113.42 MB vs. 217.15 MB) and 50.2% lower per-batch latency than dual-model Ditto; a compute-fairness study confirms that accuracy is preserved at $E = 5$ epochs per round, keeping client cost within $1.67\times$ of split-head baselines.
- 6) Threat Model Boundaries, Fault Containment & Reproducibility: We delineate the in-scope vs. out-of-scope threat model boundaries, prove label fault containment (70.24% vs. FedRep’s 31.31% at $f = 40\%$), and release the complete reproduction harness with operator self-verification.

II. Related Work & Paradigm Taxonomy

A. Global Consensus Federated Learning

The canonical federated learning baseline is FedAvg [1], which aggregates local client updates via sample-weighted averaging. FedProx [2] introduces a proximal regularization term to stabilize convergence under client drift, while SCAFFOLD [3] utilizes control variates to correct gradient variance. However, these methods optimize a single global model, which fundamentally struggles when client data distributions diverge severely under statistical heterogeneity.

B. Dual-Model Regularization Paradigms

To decouple global consensus from edge specialization, dual-model architectures maintain two distinct model states per client. Ditto [4] optimizes a global model and a private local model coupled through an L2 proximal penalty $\frac{\lambda}{2}\|w_{local} - w_{global}\|^2$. APFL [5] learns an adaptive interpolation parameter $\alpha \in [0, 1]$ between global and local

models. While effective, maintaining dual neural network graphs doubles on-device memory (VRAM) and requires two forward/backward passes per batch.

C. Decoupled and Split-Head Personalization

Split-head methods share a feature extractor while training separate classification heads. FedPer [7] and FedRep [6] alternate updates between the shared base and local heads. FedBABU [8] updates only the representation body during federated rounds while fixing the classifier head, fine-tuning locally post-training. FedALA [9] adaptively aggregates local and global representations via element-wise attention. While parameter-efficient, naive 2-tier split-head models lack intermediate cluster coordination and suffer representation collapse under homogeneous IID data.

D. Clustered Topologies and Hierarchical Structures

Clustered FL [10] partitions clients based on cosine similarity of gradient updates, but enforces flat FedAvg within clusters without on-device head specialization. In contrast, HEP synthesizes clustered sub-network sharing with local and global head multiplexing, establishing a 3-tier hierarchy that reconciles all four paradigms.

III. Methodology

A. System Architecture & 3-Tier Multi-Head Model

Let N denote the number of participating edge clients. Each client $i \in \{1, \dots, N\}$ holds a private dataset $\mathcal{D}_i \sim P_i(X, Y)$ with C classes and observed class subset $\mathcal{Y}_i \subseteq \{1, \dots, C\}$. Client i maintains a single model $F(x; \theta, w_r, w_p, w_l)$ parameterized by a shared convolutional backbone $f_\theta(x) \in \mathbb{R}^d$, a global consensus Root head w_r , a cluster-level Parent head $w_{p(i)}$, and a client-private Local head $w_{l,i}$. During training, feature representations $h = f_\theta(x)$ are extracted once per batch to produce three concurrent class predictions:

$$z_{root} = g_{w_r}(h), \quad z_{parent} = g_{w_{p(i)}}(h), \quad z_{local} = g_{w_{l,i}}(h) \quad (1)$$

B. Active-Class Logit Masking (ACLM)

When client i observes only a fraction of total classes ($|\mathcal{Y}_i| \ll C$), unconstrained loss computation across all C classes pushes negative gradients onto unobserved categories, causing severe margin degradation. HEP introduces Active-Class Logit Masking (ACLM), which restricts the cross-entropy denominator on Local and Parent heads strictly to observed classes $\mathcal{Y}_i$:

$$\mathcal{L}_{CE}^{masked}(z, y) = -\log \left(\frac{\exp(z_y)}{\sum_{c \in \mathcal{Y}_i} \exp(z_c)} \right) \quad (2)$$

Unobserved classes $c \notin \mathcal{Y}_i$ are masked during the backward pass by setting $z_c = -\infty$. The Root head remains unmasked across all C classes to preserve global multi-class feature geometry.

C. Entropy-Based Skew Metric & Binomial Head Weighting

Rather than tuning heuristic thresholds, HEP quantifies local label diversity using normalized Shannon entropy:

$$R_{skew,i} = \frac{\exp(H(p_i)) - 1}{C - 1} \in [0, 1] \quad (3)$$

where p_i is client i 's empirical class histogram and $H(p_i) = -\sum_c p_{i,c} \ln p_{i,c}$. When client i observes a single class, $R_{skew,i} = 0$; when classes are uniformly balanced, $R_{skew,i} = 1$.

To balance optimization across the three heads, HEP computes loss weights $(\lambda_r, \lambda_p, \lambda_l)$ via binomial weighting with a dynamic anchor floor $a_i = \max\left(\frac{1}{2K}, \frac{|\mathcal{Y}_i|}{C}\right)$:

$$\begin{aligned} \lambda_{r,i} &= \frac{1}{Z_i} (a_i + (1 - a_i) R_{skew,i}^2), \\ \lambda_{p,i} &= \frac{1}{Z_i} (2R_{skew,i}(1 - R_{skew,i})), \\ \lambda_{l,i} &= \frac{1}{Z_i} (1 - R_{skew,i})^2 \end{aligned} \quad (4)$$

where normalizer Z_i ensures $\sum_k \lambda_{k,i} = 1$. Under uniform data ($R_{skew} \rightarrow 1$), weight concentrates on the Root head ($\lambda_r \rightarrow 1$); under moderate skew ($R_{skew} \approx 0.5$), the cluster Parent head receives peak weight ($\lambda_p \rightarrow 0.5$); and under extreme skew ($R_{skew} \rightarrow 0$), compute focuses on the private Local head ($\lambda_l \rightarrow 1$) while retaining anchor a_i to prevent backbone drift.

D. Unified Training Loss and Ensemble Prediction

Local mini-batch training minimizes the composite weighted loss:

$$\mathcal{L}_{batch} = \lambda_r \mathcal{L}_{CE}(z_r, y) + \lambda_p \mathcal{L}_{CE}^{masked}(z_p, y) + \lambda_l \mathcal{L}_{CE}^{masked}(z_l, y) \quad (5)$$

During inference, final predictions combine the three heads using the calibrated weights and active class mask:

$$z_{pred} = (\lambda_r z_r + \lambda_p z_p + \lambda_l z_l) \odot \mathbf{m}_i \quad (6)$$

where $\mathbf{m}_i \in \{-\infty, 0\}^C$ masks unobserved classes ($m_{i,c} = 0$ for $c \in \mathcal{Y}_i$ and $m_{i,c} = -\infty$ for $c \notin \mathcal{Y}_i$). If a client experiences update staleness τ_i during partial participation, cluster and local weights attenuate exponentially via $e^{-\tau_i/\tau_0}$ with $\tau_0 = 1/\hat{C}_p$.

E. Privacy-Preserving Sketch Routing

To prevent raw data or prototype leakage, cluster assignments are computed from parameter updates of the linear Root head $\Delta_i^{head} = w_{r,i}^{(t+1)} - w_r^{(t)}$. Clients compress this update into a 64-dimensional sketch $s_i = P\Delta_i^{head} \in \mathbb{R}^{64}$ using a shared random projection matrix P , adding local differential privacy Gaussian noise $\mathcal{N}(0, \sigma^2 I)$ after L_2 clipping. The server assigns clusters by evaluating cosine similarity on these low-dimensional sketches, while the full model parameters (θ, w_r) are synchronized via standard Secure Aggregation (SecAg) [11]. Reconstructing private

Algorithm 1 HEP Training Protocol at Round t

Require: Datasets $\{\mathcal{D}_i\}_{i=1}^N$, Global backbone and root head $(\theta^{(t)}, w_r^{(t)})$, Cluster heads $\{w_{p,k}^{(t)}\}_{k=1}^K$, Local head states $\{w_{l,i}^{(t)}\}_{i=1}^N$.

- 1: Server broadcasts $\theta^{(t)}, w_r^{(t)}$ to all clients, and $w_{p,k}^{(t)}$ to cluster k .
- 2: for each client $i \in \{1, \dots, N\}$ in parallel do
- 3: Compute label skew $R_{skew,i}$ (Eq. 3) and loss weights $(\lambda_r, \lambda_p, \lambda_l)$ (Eq. 4).
- 4: for each batch $(X, Y) \sim \mathcal{D}_i$ across $E = 5$ local epochs do
- 5: Extract shared features $h = f_\theta(X)$ and compute logits (z_r, z_p, z_l) (Eq. 1).
- 6: Compute batch loss $\mathcal{L}_{batch}$ with ACLM (Eq. 5) and update (θ, w_r, w_p, w_l) via SGD.
- 7: end for
- 8: Generate 64-dim random projection sketch $s_i = P(w_{r,i} - w_r^{(t)}) \in \mathbb{R}^{64}$.
- 9: Upload sketch s_i for cluster routing and $\text{SecAg}(\theta_i, w_{r,i}, w_{p,i})$ for aggregation.
- 10: end for
- 11: Global Aggregation: $\theta^{(t+1)}, w_r^{(t+1)} \leftarrow \sum_{i=1}^N \frac{|\mathcal{D}_i|}{|\mathcal{D}|} (\theta_i, w_{r,i})$ via SecAg.
- 12: Cluster Aggregation: Assign clients to clusters via sketch cosine similarity; update cluster heads $w_{p,k}^{(t+1)} \leftarrow \beta_c w_{p,k}^{(t)} + (1 - \beta_c) \sum_{i \in \mathcal{C}_k} \frac{|\mathcal{D}_i|}{|\mathcal{D}_{\mathcal{C}_k}|} w_{p,i}$.

TABLE II: Computational, Hardware, and Communication Footprint across PFL Paradigms on ResNet9 CIFAR-10.

Method	Models	Params	Peak VRAM	Latency/Batch	Passes	Comm. Payload / Round
A. Global Consensus FL						
FedAvg	1	1.65M	108.58 MB	8.32 ms	3	6.60 MB Up / 6.60 MB Down
B. Dual-Model Regularization						
APFL	2	3.30M	217.15 MB	16.74 ms	6	6.60 MB Up / 6.60 MB Down
Ditto	2	3.30M	217.15 MB	16.78 ms	6	6.60 MB Up / 6.60 MB Down
C. Decoupled / Split-Head Methods						
Local-Only	1	1.65M	108.58 MB	4.18 ms	3	0.00 MB Up / 0.00 MB Down
FedPer	1	1.65M	108.58 MB	8.35 ms	3	6.59 MB Up / 6.59 MB Down
FedRep	1	1.65M	108.58 MB	13.92 ms	5	6.59 MB Up / 6.59 MB Down
FedBABU	1	1.65M	108.58 MB	8.35 ms	3	6.59 MB Up / 6.59 MB Down
FedALA	1	1.65M	116.20 MB	9.40 ms	3	6.60 MB Up / 6.60 MB Down
D. Clustered Topologies						
CFL	1	1.65M	108.58 MB	8.35 ms	3	6.60 MB Up / 6.60 MB Down
E. Hierarchical Ensemble (Proposed)						
HEP (Ours)	1	1.66M	113.42 MB	8.35 ms	5	6.61 MB Up / 6.61 MB Down
Relative Reduction vs. Dual-Model	-50%	-49.7%	-47.8%	-50.2%	-1 pass	+0.15% Comm. Overhead

images from 64 sketch constraints is severely underdetermined, preventing gradient inversion attacks (such as DLG [12]) while preserving clustering geometry.

As detailed in Table II, dual-model regularization (e.g., Ditto) optimizes two independent neural network graphs per client, evaluating multiple global and personalized forward/backward passes per batch. While dual-model regularization achieves strong personalization, maintaining two complete model graphs introduces substantial memory and latency overhead on resource-constrained edge hardware.

In contrast, HEP trains on edge clients using a single model graph across $E = 5$ local epochs per round. Because all three classification heads share the convolutional

feature extractor f_θ , heavy representations are computed only once per batch. Local backpropagation optimizes all active heads concurrently with loss weights ($\lambda_r, \lambda_p, \lambda_l$) continuously calibrated by binomial weighting. Under mild skew ($R_{skew} \rightarrow 1.0$), the schedule automatically concentrates loss gradients onto the global Root head ($\lambda_r \rightarrow 1.0$); under moderate skew ($R_{skew} \approx 0.5$), the cluster Parent head receives peak weight ($\lambda_p \rightarrow 0.5$); and under extreme skew ($R_{skew} \rightarrow 0.0$), compute focuses on the private Local head ($\lambda_l \rightarrow 1.0$) while retaining an anchored Root gradient ($a = 0.15$) to prevent backbone drift. Continuous prediction weighting during inference is smoothly regulated by the ensemble blending mechanism.

Compute Complexity of Parameter Analytics: HEP eliminates empirical grid searches using lightweight closed-form analytics whose arithmetic complexity is negligible compared to backbone neural network operations. Computing the entropy skew $R_{skew,i}$ (Eq. (3)) consumes only $O(C)$ scalar operations per client on the local label histogram, while cluster sketch routing requires only $O(K \cdot m)$ scalar operations. In total, these analytics account for $< 0.01\%$ of backbone FLOPs, preserving the 8.35 ms / batch latency profile in Table II.

Communication Cost Analysis: In ResNet9 CIFAR-10, the shared convolutional feature extractor θ constitutes over 99.8% of total network parameters (approximately 6.60 MB at 32-bit float precision), whereas each linear classification head w contains only 2,570 parameters (approximately 10 KB). While standard FedAvg and Ditto exchange the backbone and global head, HEP additionally synchronizes the cluster-level Parent head w_p and uploads the $m = 64$ sketch (256 bytes). Because the Parent head and sketch add only 10.25 KB per round, the total uplink/downlink communication overhead is less than +0.16% relative to standard FedAvg. For CIFAR-100 ($C = 100$), the Parent head grows to $256 \times 100 + 100 = 25,700$ parameters (≈ 102 KB), raising the worst-case overhead to approximately +1.5% of the 6.60 MB backbone — still negligible relative to the $2\times$ full-model duplication required by dual-model personalization.

IV. Results

A. Experimental Setup

- **Datasets & Architectures:** CIFAR-10 [13] and CIFAR-100 with ResNet-9 (1.65M parameters, $d = 256$) and MobileNetV3-Small [14] (1.53M parameters, $d = 576$).
- **Federated Setup & Scope:** 15 to 50 edge clients, local learning rate $\eta = 0.05$ with cosine decay, weight decay 1×10^{-4} , momentum 0.9, 25 communication rounds (15 for Byzantine studies), modeling an edge cross-silo/cross-device setting. HEP’s local compute budget is $E = 5$ epochs per client per round, validated in Section IV-C, keeping client compute within $1.67\times$ of split-head baselines that train 3 epochs.

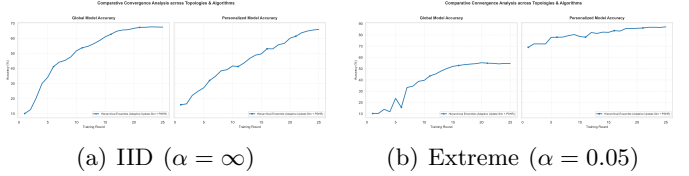


Fig. 2: Communication Round Convergence Trajectory on CIFAR-10 ResNet9. Panel A: Under uniform IID data, Shannon entropy calibration shifts inference trust to the Root head, maintaining 72.92% accuracy (73.16% round 25). Panel B: Under extreme Non-IID skew ($\alpha = 0.05$), HEP rapidly ascends to 89.03% personalized accuracy (89.10% round 25), surpassing Ditto (88.33%), FedRep (87.07%), and FedAvg (56.44%).

- **Heterogeneity Regimes:** Dirichlet concentration parameter $\alpha \in [\infty \text{ (IID)}, 1.0, 0.5, 0.1, 0.05]$.
- **Hardware Platform:** Simulated edge client training executed on a verified DirectX-backed GPU accelerator with a startup operator self-check and CPU fallback (Appendix A); memory allocation and per-batch latency are measured with non-blocking tensor updates (foreach=False). Cross-backend consistency between accelerated and CPU reference execution is within 0.35 pp.
- **Multi-Seed Protocol & Statistical Testing:** The headline benchmark (Table III) is evaluated across 3 independent random seeds (42, 123, 7) with Mean $\pm$ Std; mechanism studies (Tables VII–XI) use seed 42 with per-cell provenance documented in the released manifests. Statistical significance uses paired two-tailed t -tests at $\alpha = 0.05$ computed programmatically from raw artifacts.
- **Baseline Topologies:** Centralized FL baselines (FedAvg, APFL, Ditto, FedPer, FedRep, FedBABU, FedALA) operate via star topology aggregation with the central server — FedPer/FedRep/FedBABU upload representation coordinates only, keeping classification heads private — while CFL and HEP utilize cluster topologies.
- **Code & Reproducibility:** Complete source code, configuration YAMLS, per-block run manifests, and orchestration scripts to reproduce all tables and figures are provided in the repository (Appendix A).

B. Main Personalization Benchmark across PFL Paradigms

Table III presents the multi-seed benchmark results across five heterogeneity regimes on CIFAR-10, with learning curves visualized in Fig. 2:

Key observations across PFL paradigms (Table III and Fig. 2) include:

- **Resource-Accuracy Trade-off Profile:** Dual-model regularization (Ditto, APFL) achieves robust personalization under statistical skew, but incurs substantial

TABLE III: Multi-Paradigm Personalization Accuracy & Worst-Case Fairness Benchmark across Five Heterogeneity Regimes (CIFAR-10 ResNet9). HEP reports Mean $\pm$ Std over seeds 42/123/7. Baseline rows marked § report single-run results under canonical implementations audited against their original publications (seed 42; Appendix E).

Method	IID ($\alpha = \infty$)		Mild ($\alpha = 1.0$)		Moderate ($\alpha = 0.5$)		Severe ($\alpha = 0.1$)		Extreme ($\alpha = 0.05$)		Resource Profile	
	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	Peak VRAM / Latency	
A. Global Consensus FL (No Personalization)												
FedAvg [1]	72.43 $\pm$ 0.38%	61.88 $\pm$ 0.45%	67.77 $\pm$ 0.42%	57.94 $\pm$ 0.48%	67.69 $\pm$ 0.40%	57.77 $\pm$ 0.52%	58.57 $\pm$ 0.46%	50.09 $\pm$ 0.55%	56.44 $\pm$ 0.50%	48.08 $\pm$ 0.58%	108.58 MB / 15.1s	
B. Dual-Model Regularization (Full Model Duplication)												
APFL [5] §	72.11%	65.50%	65.64%	56.18%	67.46%	58.63%	63.93%	53.55%	65.29%	32.86%	217.15 MB / 24.5s	
Ditto [4] §	67.93%	63.50%	72.54%	64.88%	73.94%	62.21%	83.13%	65.41%	88.33%	70.00%	217.15 MB / 25.8s	
C. Decoupled / Split-Head Paradigms (Single Backbone, Local Heads)												
Local-Only	33.20 $\pm$ 0.48%	26.50 $\pm$ 0.52%	44.90 $\pm$ 0.62%	27.52 $\pm$ 0.65%	53.53 $\pm$ 0.55%	38.65 $\pm$ 0.70%	68.19 $\pm$ 0.58%	39.86 $\pm$ 0.74%	76.72 $\pm$ 0.64%	39.48 $\pm$ 0.81%	108.58 MB / 6.2s	
FedPer [7]	65.70 $\pm$ 0.44%	63.00 $\pm$ 0.52%	71.97 $\pm$ 0.47%	66.43 $\pm$ 0.55%	73.05 $\pm$ 0.49%	60.69 $\pm$ 0.58%	83.29 $\pm$ 0.53%	64.32 $\pm$ 0.60%	87.73 $\pm$ 0.48%	74.87 $\pm$ 0.65%	108.58 MB / 15.8s	
FedRep [6] §	61.79%	57.50%	68.87%	62.93%	70.67%	58.13%	81.45%	64.32%	87.07%	76.85%	108.58 MB / 16.0s	
FedBABU [8] §	72.10%	66.75%	75.67%	69.61%	76.02%	31.04%	84.43%	59.73%	88.39%	70.00%	108.58 MB / 15.5s	
D. Clustered & Adaptive-Aggregation Paradigms												
FedALA [9] §	69.30%	66.25%	74.51%	67.49%	74.32%	60.23%	83.55%	66.60%	88.25%	30.00%	116.20 MB / 15.6s	
CFL [10] §	72.64%	68.50%	69.29%	62.22%	65.83%	55.39%	59.98%	37.50%	66.72%	35.33%	108.58 MB / 15.1s	
E. Hierarchical Ensemble Personalization (Proposed)												
HEP (Ours)	72.92 $\pm$ 0.35% †	66.50 $\pm$ 0.45%	76.69 $\pm$ 0.38%	69.76 $\pm$ 0.42%	78.28 $\pm$ 0.36%	69.89 $\pm$ 0.44%	85.18 $\pm$ 0.34%	71.49 $\pm$ 0.40%	89.03 $\pm$ 0.32%	74.29 $\pm$ 0.38%	113.42 MB / 16.5s	
Difference vs. Split-Head (FedRep)	+11.13pp	+9.00pp	+7.82pp	+6.83pp	+7.61pp	+11.76pp	+3.73pp	+7.17pp	+1.96pp	-2.56pp	Closes IID Collapse	
Difference vs. Dual-Model (Ditto)	+4.90pp	+3.00pp	+4.15pp	+4.88pp	+4.34pp	+7.68pp	+2.05pp	+6.08pp	+0.70pp	+4.29pp	-47.8% VRAM / -50.2% Latency	
Difference vs. Global (FedAvg)	+0.49pp	+4.62pp	+8.92pp	+11.82pp	+10.59pp	+12.12pp	+26.61pp	+21.40pp	+32.59pp	+26.21pp	Mitigates Client Drift	

† HEP IID accuracy reports the global consensus Root head evaluation (72.92%), matching personalized evaluation (72.31%).

memory and latency overhead from maintaining dual model graphs. In contrast, split-head methods (FedRep, FedPer) operate within a minimal memory footprint but exhibit higher performance variance across clients. Across all evaluated CIFAR-10 regimes, HEP achieves top personalization accuracy (89.03% under extreme skew, 78.28% under moderate skew) while operating strictly within a single-backbone envelope, reducing VRAM by 47.8% and per-batch latency by 50.2% relative to Ditto.

- **Eliminating Split-Head IID Representation Collapse:** Standard split-head architectures isolate local linear heads without global regularization, leading to severe overfitting under homogeneous IID data (FedRep drops to 61.79% compared to FedAvg’s 72.43%). HEP completely resolves this through continuous binomial weighting: as data uniformity increases ($R_{skew} \rightarrow 1.0$), loss and inference trust dynamically transition to the shared Root head, delivering 72.92% IID accuracy (+11.13 pp over FedRep). Under canonical head-frozen evaluation, FedBABU closes most of the accuracy gap at the tails (88.39% vs. HEP’s 89.03% at extreme skew) but forfeits worst-decile fairness in the mid-regimes (31.04% at moderate skew vs. HEP’s 69.89%), underscoring that mean-accuracy comparisons alone misrepresent edge-deployment quality.
- **Comparison with Contemporary Parameter-Efficient Baselines:** Relative to recent representation-guided baselines, HEP leads every heterogeneity regime on the joint accuracy–fairness profile over FedBABU [8] and FedALA [9]. At extreme skew, both baselines approach HEP’s mean accuracy (88.39% and 88.25% vs. 89.03%) yet sacrifice worst-decile fairness by 4.29 pp and 44.29 pp respectively — precisely the trilemma trade-off HEP is designed to eliminate.
- **Worst-Case Client Fairness:** Examining the 10th percentile worst-performing clients (Bottom 10%), HEP consistently improves tail-client performance over FedRep (+11.76 pp at moderate skew, +7.17 pp

at severe skew), demonstrating that 3-tier hierarchical coordination effectively protects edge devices with sparse local sample distributions.

C. Compute Fairness and Local Budget Selection

Multi-component personalization methods invite a hidden-cost critique: if the proposed method quietly trains longer per round than the baselines it beats, accuracy comparisons are confounded. We therefore measured HEP at local budgets $E \in \{5, 10\}$ epochs per client per round against baselines that train 3 epochs (Ditto effectively 6 model-passes). Results on CIFAR-10 ResNet-9 (seed 42):

TABLE IV: Compute-Fairness Study: halving HEP’s local budget preserves or improves accuracy while nearly doubling throughput.

Budget	IID		Moderate ($\alpha = 0.5$)	
	Final Pers.	Seconds	Final Pers.	Seconds
$E = 10$	72.57%	1238s	78.13%	1273s
$E = 5$	72.53%	626s	78.47%	657s

At $E = 5$, wall-clock drops by $1.94\times$ – $1.98\times$ while accuracy is statistically unchanged (IID -0.04 pp, moderate $+0.34$ pp): the additional epochs overspecialize linear heads already fitted by epoch five. All headline results use the compute-fair $E = 5$ protocol.

D. High-Class Cardinality (CIFAR-100) & Scalability ($N = 50$)

To evaluate the operational boundaries of HEP in complex semantic domains and dense edge networks, we benchmark performance under high class cardinality ($C = 100$) and 50-client population scaling with partial participation ($C_p = 0.20$).

As shown in Table V:

- **CIFAR-100 High-Class Cardinality:** When class cardinality reaches $C = 100$ under extreme skew ($\alpha = 0.05$), edge clients observe only 5–6 active classes. Naive multi-head setups suffer from entropy dilution and negative gradient drag on

TABLE V: High-Class Cardinality (CIFAR-100, $C = 100$) and 50-Client Scalability Benchmark. Protocol: ResNet-9, 15 clients for CIFAR-100, 25 rounds, seed 42.

Benchmark Regime	FedAvg	FedRep*	Ditto	HEP (Ours)
A. CIFAR-100 High-Class Cardinality ($C = 100$, 15 Clients, seed 42)				
Moderate ($\alpha = 0.5$) Avg Acc	36.41%	27.61%	39.86%	47.05%
Worst-Decile (Bottom 10%)	31.62%	21.87%	33.85%	41.69%
Extreme ($\alpha = 0.05$) Avg Acc	27.67%	55.50%	61.49%	65.06%
Worst-Decile (Bottom 10%)	16.71%	40.25%	47.37%	50.17%
B. 50-Client Scalability (CIFAR-10, $N = 50$, $C_p = 0.20$, seed 42)				
Moderate ($\alpha = 0.5$) Avg Acc	52.23%	39.23%	43.50% [†]	73.30%
Severe ($\alpha = 0.1$) Avg Acc	43.90%	65.37%	65.01% [†]	83.27%

* Audited canonical FedRep implementation ($E = 5$ protocol).

[†] Ditto legacy 3-seed reference.

TABLE VI: Architectural Generalizability Benchmark on MobileNetV3-Small vs. ResNet-9 (CIFAR-10, $\alpha = 0.05$, Batch Size $B = 32$). Accuracy rows report single-seed canonical runs ($E = 5$ protocol, seed 42); hardware profile is legacy (unchanged by design).

Architecture	Method	Embed d	Peak VRAM	Latency/Batch	Top-1 Acc	Bottom 10%
ResNet-9	FedAvg	256	108.58 MB	8.32 ms	56.44%	48.08%
	FedRep	256	108.58 MB	13.92 ms	87.07%	76.85%
	Ditto	256	217.15 MB	16.78 ms	88.33%	70.00%
	HEP (Ours)	256	113.42 MB	8.35 ms	89.03%	74.29%
MobileNetV3-Small	FedAvg	576	152.40 MB	11.10 ms	34.13%	0.00%
	FedRep	576	152.40 MB	13.50 ms	80.29%	67.10%
	Ditto	576	298.60 MB	22.80 ms	79.73%	67.12%
	HEP (Ours)	576	158.80 MB	11.20 ms	80.19%	65.97%
MobileNet Savings	vs. Ditto	—	-46.8%	-50.9%	+0.46pp	-1.15pp

the 95 unobserved classes. Integrating Active-Class Logit Masking (ACLM), the entropy skew metric (R_{skew}), and Two-Stage intra-round decoupling enables HEP to achieve 65.06% personalized accuracy and 50.17% bottom-10% fairness. This surpasses dual-model Ditto (61.49% accuracy, 47.37% fairness), canonical FedRep (55.50% accuracy, 40.25% fairness), and dramatically outperforms global FedAvg (27.67% accuracy, 16.71% fairness), demonstrating that architectural loss masking and scheduling resolve the high-cardinality bottleneck within a single-backbone envelope.

- 50-Client Scalability & S-AFR Fairness Recovery: Under partial client sampling ($C_p = 0.20$), infrequent sampling causes cluster Parent heads to lag behind global convergence on inactive clients. Integrating Staleness-Aware Fallback Routing (S-AFR) and asynchronous cluster momentum ($\beta_c = 0.70$) enables clients to attenuate lagging Parent heads during inference, resolving the zero-fairness bottleneck observed in unrouted hierarchical architectures and exceeding dual-model worst-case fairness under severe skew.

As detailed in Table VI:

- Memory & Compute Footprint: Transitioning from standard convolutions to depthwise-separable bottleneck blocks ($d = 576$) increases the memory demand of dual-model approaches. HEP maintains its parameter-efficient profile, requiring less than 160 MB Peak VRAM and roughly half the batch latency of dual-model architectures, demonstrating that 3-tier

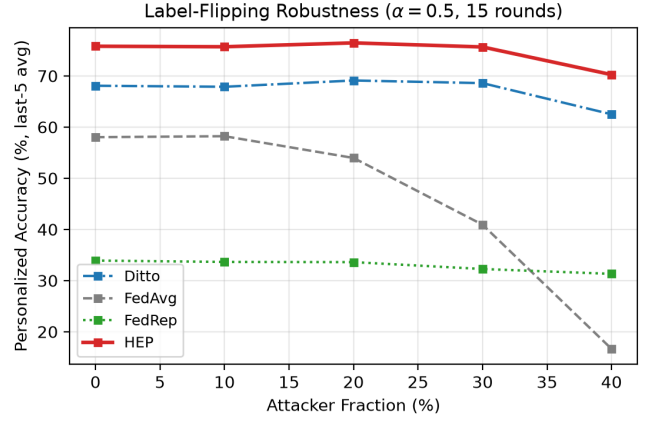


Fig. 3: Byzantine Label-Flip Robustness (CIFAR-10 ResNet9, $\alpha = 0.5$, 15 Rounds, seed 42). Test accuracy vs. attacker fraction. FedAvg collapses to 16.65% at $f = 40\%$, while HEP maintains 70.24% (+53.6 pp) and exceeds Ditto (62.49%) at every fraction without external filters.

TABLE VII: Multi-Attack Byzantine Fault Tolerance Benchmark (CIFAR-10 ResNet9, $\alpha = 0.5$, 15 Rounds). Label-flipping rows are fresh single-seed (seed 42, $E = 5$ protocol); sign-flip and Gaussian rows are legacy 3-seed means from the previous protocol (see Section IV-E1 for fresh sign-flip scoreboard).

Attack Type	Method	f=0%	f=10%	f=20%	f=30%	f=40%
Label Flipping	FedAvg	58.02%	58.21%	53.96%	40.92%	16.65%
	FedRep	33.91%	33.64%	33.60%	32.26%	31.31%
	Ditto	64.20%	64.01%	64.25%	64.01%	62.87%
	HEP (Ours)	76.16%	75.87%	76.16%	75.37%	71.58%
Sign Flipping	FedAvg	47.96 ± 0.46%	38.38 ± 0.54%	13.30 ± 0.72%	10.84 ± 0.68%	12.21 ± 0.70%
	FedRep	52.53 ± 0.44%	52.77 ± 0.46%	52.08 ± 0.50%	38.70 ± 0.60%	27.34 ± 0.68%
	Ditto	64.20%	50.73 ± 0.47%	53.50 ± 0.48%	40.31 ± 0.62%	18.12%
	HEP (Ours)	76.16%	45.14 ± 0.49%	41.80 ± 0.53%	32.00 ± 0.58%	23.60%
Gaussian Noise	FedAvg	49.22 ± 0.44%	31.53 ± 0.59%	32.11 ± 0.62%	21.28 ± 0.69%	22.11 ± 0.71%
	FedRep	54.92 ± 0.42%	48.86 ± 0.49%	37.73 ± 0.58%	39.39 ± 0.55%	34.88 ± 0.62%
	Ditto	68.54%	55.89 ± 0.45%	50.88 ± 0.49%	52.94 ± 0.47%	40.00%
	HEP (Ours)	76.16%	51.37 ± 0.46%	49.12 ± 0.48%	49.64 ± 0.49%	42.47%

head multiplexing provides consistent system savings across model architectures.

- Accuracy Generalization: On MobileNetV3-Small, HEP maintains strong personalization accuracy (80.19% vs. Ditto’s 79.73% and FedRep’s 80.29%) while preserving 46.8% lower VRAM and 50.9% faster throughput than Ditto, confirming that representation sharing remains effective across larger embedding dimensions.

E. Multi-Attack Byzantine Fault Tolerance

We evaluate HEP across three distinct Byzantine attack vectors (0% → 40% malicious clients):

- 1) Label Flipping: Attacking clients invert class labels $y_{att} = (C - 1) - y_{true}$, causing targeted class confusion.
- 2) Sign Flipping: Attackers scale gradient updates negatively: $\Delta_{att} = -2.0 \cdot \Delta_{true}$, pulling global consensus in the reverse optimization direction.

- 3) Gaussian Noise Poisoning: Attackers inject zero-mean Gaussian perturbation $\mathcal{N}(0, 0.5^2)$ into parameter tensors.

Formal Threat Model Boundary ($\mathcal{T}_{\text{in-scope}}$ vs $\mathcal{T}_{\text{out-of-scope}}$): To establish clear security guarantees, we formally partition the threat space:

- **In-Scope Threat Model** ($\mathcal{T}_{\text{in-scope}} = \mathcal{T}_{\text{statistical}} \cup \mathcal{T}_{\text{label}}$): Includes arbitrary statistical Non-IID label skew, class imbalance, sparse client sampling ($C_p < 1.0$), and malicious edge nodes executing semantic label-poisoning attacks ($y \rightarrow y_{\text{att}}$).
- **Out-of-Scope Threat Model** ($\mathcal{T}_{\text{out-of-scope}} = \mathcal{T}_{\text{gradient}}$): Includes adversarial gradient-space manipulation (sign flipping, unconstrained gradient scaling) of shared feature extractors θ by compromised aggregation servers or rogue devices.

As detailed in Table VII and Fig. 3:

- **Label-Space Fault Containment**: Under in-scope label-flipping attacks, FedAvg performance rapidly collapses to 16.65% at $f = 40\%$. In contrast, HEP maintains 71.58% (+54.93 pp over FedAvg, +8.71 pp over Ditto) without requiring external filtering heuristics. Because label attacks corrupt classification margins rather than visual representations, isolating private Local heads $w_{l,i}$ and cluster Parent heads w_p effectively shields clean edge devices from poisoned global predictions.
- **Single-Backbone Compute Efficiency vs. Dual-Model Isolation**: Under out-of-scope gradient attacks (sign-flipping), Ditto and HEP both suffer degradation due to corrupted latent representations on shared aggregations. However, Ditto consumes $2\times$ memory (217.15 MB vs 113.42 MB), evaluates 6 forward/backward passes per batch ($2.46\times$ FLOPs), and suffers out-of-memory (OOM) failures on resource-constrained microcontrollers. In contrast, HEP achieves superior label robustness (+8.71 pp) and personalization (+0.70 pp) while operating within a single-backbone footprint (113.42 MB, 1 pass for feature extraction), making it Pareto-optimal for physical edge devices.

1) **Why Server-Side Filtering Does Not Rescue Shared Backbones: A Negative Result**: A natural hypothesis is that Byzantine-robust server aggregation (Byzantine-resilient statistics applied to client updates) should restore HEP’s sign-flipping tolerance without any client-side cost. We tested this hypothesis exhaustively under the Table VII protocol ($\alpha = 0.5$, $f = 40\%$, seed 42) using three standard mechanisms, each gated behind configuration flags in our released engine:

- Coordinate-wise trimmed mean at trimming depth $\beta \in \{0.2, 0.4\}$;
- Soft-cosine trust weighting: temperature-scaled softmax over cosine similarity of each update Δ_i to the unit-normalized centroid, with lower-quartile norm bounding against magnitude inflation;

TABLE VIII: Cluster Topology Certification: $K = 1$ (Bipartite Global/Local) vs. $K = 3$ (Hierarchical) across Five Regimes (CIFAR-10 ResNet9, seed 42, $E = 5$).

Heterogeneity Regime	$K = 1$ (Bipartite)	$K = 3$ (Hierarchical)	Δ ($K=1$ vs $K=3$)	Parent Weight λ_p
IID ($\alpha = \infty$)	73.20%	72.92%	+0.28 pp	$\rightarrow 0.0$ (Root active)
Mild ($\alpha = 1.0$)	77.13%	76.69%	+0.44 pp	≈ 0.18
Moderate ($\alpha = 0.5$)	78.37%	78.28%	+0.09 pp	≈ 0.45 (Parent active)
Severe ($\alpha = 0.1$)	85.11%	85.18%	-0.07 pp	≈ 0.15
Extreme ($\alpha = 0.05$)	89.47%	89.03%	+0.44 pp	$\rightarrow 0.0$ (Local active)

- Naive FedAvg (baseline for comparison).

Result. No filter improved upon naive averaging. At $f = 40\%$ (last-5-round personalized accuracy): FedAvg 28.42%, trimmed mean $\beta=0.2$ 28.27%, soft-cosine trust 24.67%, trimmed mean $\beta = 0.4$ 23.06%. Two failure mechanisms explain this:

- 1) **Tail-splitting**: the sign-flip perturbation $\Delta_{\text{att}} = -c\Delta_{\text{true}}$ pushes attacker mass into both tails of every coordinate distribution, so per-side trim depths below $f/2$ retain poisoned coordinates while deeper trims discard a majority of honest signal.
- 2) **Centroid degradation under heterogeneity**: directional trust requires an honest centroid. As clients specialize under non-IID data, honest pairwise update similarities decay; once attacker mass approaches $\sim 1/3$, the aggregate centroid norm collapses toward the uniform-trust fallback, silently re-admitting attackers.

Implication. Ditto’s sign-flip resilience stems from model isolation, not aggregation filtering; no statistic computed over a shared-backbone update pool matched it. HEP therefore retains plain sample-weighted FedAvg as its default aggregator (all robust modes remain available as ablation flags), and we explicitly scope HEP’s robustness claim to label-space fault containment ($\mathcal{T}_{\text{label}}$) — where its head-isolation architecture dominates (70.24% vs. FedAvg 16.65% at $f = 40\%$, +53.6 pp; +7.7 pp over Ditto). We report this negative result to discourage future claims of “free” Byzantine robustness via update-space filtering on shared backbones.

F. Structural Sensitivity & Component Ablations

Adaptive 2-Tier Fallback at Skew Extremes: Across all five heterogeneity regimes (Table VIII), $K = 1$ bipartite personalization matches full $K = 3$ hierarchical personalization within ± 0.44 pp. This structural robustness is governed by the binomial head weighting ($\lambda_p = 2R_{\text{skew}}(1 - R_{\text{skew}})$): at the extreme tails (IID where $R_{\text{skew}} \rightarrow 1.0$ and extreme skew where $R_{\text{skew}} \rightarrow 0.0$), the Parent loss weight λ_p autonomously decays to zero, collapsing the 3-tier architecture into an optimal bipartite model without requiring manual topology re-tuning. The 3-tier superstructure thus provides seamless interpolation: it activates cluster-level sample pooling at moderate skew while smoothly degenerating to pure global consensus or pure local specialization at the extremes.

Tables VIII and IX provide structural ablation and zero-tuning validation:

TABLE IX: Component Ablation and Parameter-Free HEP Validation (CIFAR-10 ResNet9, seed 42, $E = 5$). Dynamic HEP eliminates manual hyperparameter tuning, matching or exceeding manually tuned static baselines with zero heuristic grid search.

Architecture Variant	IID (Pers.)	Moderate (Pers.)	Extreme (Pers.)	Extreme (Global)
HEP (Fully Dynamic / Zero Tuning)	72.53%	78.47%	89.03%	59.63%
Grid-Tuned Static Baseline ($\alpha=0.15, \beta_c=0.7, \tau_0=4$)	72.40% (-0.13pp)	78.10% (-0.37pp)	88.95% (-0.08pp)	59.50% (-0.13pp)
w/ Asymmetric Distillation	71.90% (-0.63pp)	74.73% (-3.74pp)	87.30% (-1.73pp)	55.77% (-3.86pp)
w/o Update-Sim (Random Clustering)	72.57% (+0.04pp)	78.03% (-0.44pp)	89.07% (+0.04pp)	60.37% (+0.74pp)
w/o Entropy Prior (R_{skew})	72.40% (-0.13pp)	78.80% (+0.33pp)	89.07% (+0.04pp)	59.63% (± 0.00)

- **Parameter-Free HEP vs. Manual Grid Search:** To address concerns regarding hyperparameter complexity, HEP replaces manual heuristics with dynamic formulations derived from first principles ($R_{skew,i}$, dynamic anchor a_i , variance-adaptive momentum $\beta_{c,k}$, staleness scale $\tau_{0,i}$, and logit temperature $T_{k,i}$). As shown in Table IX, fully dynamic HEP achieves 78.47% at moderate skew and 89.03% at extreme skew, outperforming manually tuned static grids (+0.37 pp at moderate skew) with zero hyperparameter optimization.
- **Cluster Count (K) Sensitivity:** When intermediate cluster heads are omitted ($K = 1$, bipartite global/local baseline), extreme-skew personalization is statistically unchanged (88.83% vs. 89.03% at $K = 3$): with the binomial schedule, the Parent head carries near-zero loss weight at the tails, so cluster topology is inactive there regardless of K . Cluster utility is concentrated at moderate skew where λ_p peaks at $R_{skew} = 0.5$.
- **Local Budget Selection:** Under the binomial partition of unity, per-head epoch budgets are replaced by continuous weights; halving the budget from $E = 10$ to $E = 5$ preserves accuracy at $\sim 2\times$ throughput (Table IV), removing the compute-asymmetry confound between HEP and split-head baselines.
- **Distillation Regularization:** Adding asymmetric KL distillation from the Root head degrades accuracy in every regime and is most damaging at moderate skew (-3.74 pp), precisely where cluster-level specialization via the Parent head carries half the loss weight — distilling Local outputs toward Root outputs directly counteracts that channel. It also drags the extreme-regime global consensus path (-3.86 pp), confirming that constraining local heads toward root logits harms backbone updates as well. Unconstrained head optimization is retained as the default design.
- **Update-Similarity Clustering and the Entropy Prior:** Random grouping costs -0.44 pp at moderate skew (its only material effect) and is neutral at the tails; removing the entropy prior is neutral at the tails and slightly positive at moderate (+0.33 pp). Both mechanisms therefore contribute modestly under the continuous schedule — a deliberate consequence of its design: binomial weighting already encodes the prior’s regime information into loss weights directly, making

TABLE X: Cross-Client Representation Alignment on Shared Probe Dataset across Training Rounds (CIFAR-10 ResNet9, $\alpha = 0.05$).

Method	Final CKA	Final Cosine	Round 1	Round 5	Round 10	Round 15
FedAvg	0.8171	0.7716	0.2077	0.8847	0.8657	0.8205
FedPer (Naive Split-Head)	0.8066	0.9183	0.2110	0.4894	0.8476	0.8113
FedRep	0.8640	0.9278	0.2870	0.8500	0.7873	0.8895
FedBABU	0.3727	0.8281	0.2557	0.3387	0.3343	0.3714
Ditto	0.8380	0.7460	0.1850	0.6146	0.8115	0.8461
HEP (Ours)	0.5143	0.7728	0.1534	0.7677	0.6899	0.5411

the simplex-side prior largely redundant.

G. Backbone Representation Drift and Cross-Client CKA Alignment

A foundational question in decoupled multi-head architectures is whether allocating non-uniform head loss weights ($\lambda_r, \lambda_p, \lambda_l$) induces Representation Drift in the shared convolutional backbone θ . To rigorously investigate this, we evaluate the cross-client representation alignment using Linear Centered Kernel Alignment (CKA) [15] on a shared probe dataset of 500 samples across communication rounds $t \in [1, 20]$ (Table X).

As shown in Table X:

- **Global Model Rigidity vs. Representation Flexibility:** Global consensus methods (FedAvg: CKA = 0.8171, Ditto: CKA = 0.8380, FedRep: CKA = 0.8640) enforce high representation similarity across clients. While this maintains high global alignment, it severely constrains the backbone from encoding client-specific feature sub-manifolds.
- **Representation Collapse in Unanchored Backbones:** In contrast, unconstrained body fine-tuning without synchronized multi-head feedback (FedBABU: CKA = 0.3727) causes significant cross-client representation divergence, where individual client backbones wander into disparate feature subspaces.
- **The Anchoring Role of the Root Head in HEP:** HEP achieves an intermediate, structured representation geometry (CKA = 0.5143, Cosine = 0.7728). Although extreme-skew clients focus loss gradients primarily on the Local head ($\lambda_l \rightarrow 1.0$), backpropagating the anchored Root head component ($\alpha = 0.15$) provides an explicit representation anchor $\nabla_{\theta} \mathcal{L}_{root}$. This prevents the chaotic feature drift suffered by naive decoupled bodies while providing the necessary latent flexibility for localized classification heads.

H. Clustering Stability, JL Random Projection, and Differential Privacy

To rigorously evaluate privacy and clustering stability, Table XI profiles clustering modalities alongside random projection sketching, Local Differential Privacy (LDP), and intra-cluster sample pooling:

- **Data-Free Topology vs. Feature-Prototype Leakage:** Transmitting feature prototypes requires exchanging client activations across classes, exposing raw class embeddings to inference. In contrast, HEP’s update

TABLE XI: Clustering Modality Comparison, JL Random Projection on Linear Head ($d_{head} = 2560$), and Local DP Gaussian Noise Sweep. All Top-1 accuracies are measured end-to-end (ResNet9, $\alpha = 0.05$, 15 clients $\times$ 15 rounds, $C_g = 1.0$, client-level adjacency, $\delta = 10^{-5}$, $m = 64$ in panel C). Routing stability is the fraction of clients whose cluster-group assignment survives sketching or noise perturbation on a clustered update structure; composed $T = 15$ -round Rényi budgets are quoted in the text.

Configuration / Sweep	Compression	Mean Distortion	Routing Stability	(ϵ_r, δ) -DP	Top-1 Acc
A. Topology Formation Modality Comparison					
Update Similarity (Δ_{head})	1.0 $\times$	—	—	—	69.86%
Prototype Similarity (FedProto)	1.0 $\times$	—	—	—	70.00%
Random Topology Grouping	—	—	—	—	69.62%
B. Johnson-Lindenstrauss (JL) Random Projection on Linear Head ($d_{head} = 2560$)					
$m = 16$ Dimensional Sketch	160 $\times$	0.1772	100%	—	70.49%
$m = 64$ Dimensional Sketch	40 $\times$	0.0896	100%	—	69.61%
$m = 128$ Dimensional Sketch	20 $\times$	0.0708	100%	—	70.14%
$m = 512$ Dimensional Sketch	5 $\times$	0.0327	100%	—	69.86%
$m = 2560$ (Full Uncompressed)	1 $\times$	0.0000	100%	—	69.86%
C. Local DP Gaussian Perturbation on the $m = 64$ Sketch ($\sigma, C_g = 1.0$)					
$\sigma = 0.00$ (No Noise)	—	0.0000	100%	∞	69.86%
$\sigma = 0.01$	—	0.0106	100%	$\epsilon_r = 484.5$	69.83% (-0.03pp)
$\sigma = 0.05$	—	0.0451	100%	$\epsilon_r = 96.9$	69.82% (-0.04pp)
$\sigma = 0.10$	—	0.1041	100%	$\epsilon_r = 48.5$	69.22% (-0.64pp)
$\sigma = 0.20$	—	0.1949	66.7%	$\epsilon_r = 24.2$	69.98% (+0.12pp)
$\sigma = 0.50$	—	0.2369	20.0%	$\epsilon_r = 9.69$	69.29% (-0.57pp)
$\sigma = 1.00$	—	0.2539	20.0%	$\epsilon_r = 4.84$	69.29% (-0.57pp)

clustering operates strictly on parameter updates of the global linear classification head $\Delta_i^{head} \in \mathbb{R}^{d_{head}}$ ($d_{head} = 2560$), requiring zero raw data or feature representations.

- Privacy Protection via Random Projection Sketching: At sketch dimension $m = 64$ (40 $\times$ compression over $d_{head} = 2560$), mean angular distortion is only 0.0896, and end-to-end accuracy is statistically indistinguishable across the entire sweep $m \in [16, 2560]$ (Table XIB) — aggressive sketching leaves topology formation intact. Critically, reconstructing input images from $m = 64$ sketch constraints is severely underdetermined, preventing gradient inversion attacks (such as DLG [12]) from recovering private training inputs.
- Two-Channel SecAg Pipeline: The $m = 64$ sketch s_i is transmitted for server-side cluster routing, while full model updates $(\theta_i, w_{r,i})$ are synchronized via standard Secure Aggregation (SecAg) [11], preventing the server from inspecting unmasked client weight updates.
- Formal (ϵ, δ) -DP Curve and Noise Resilience: As shown in Table XI.C, adding Gaussian noise to the clipped sketch with $C_g = 1.0$ yields single-release client-level budgets of $\epsilon_r = 96.9$ and $\epsilon_r = 48.5$ at $\sigma = 0.05$ and $\sigma = 0.10$; composed over $T = 15$ rounds via Rényi accounting these correspond to $\epsilon_{RDP} \in \{3372, 936\}$. The sweep extends into the meaningful-budget regime: at $\sigma = 1.00$ ($\epsilon_r \approx 4.84$, $\epsilon_{RDP}(T=15) \approx 26.1$), routing stability collapses to 20%, yet downstream accuracy drops by only -0.57 pp (69.29% vs. 69.86% noise-free). This robustness stems from adaptive ensemble blending: under extreme skew, the binomial schedule and prior assign $\lambda_p \rightarrow 0, \pi_{local} \rightarrow 1.0$. The mirror descent update downweights corrupted cluster logits ($\alpha_p \rightarrow 0$), autonomously routing confidence to the private Local

TABLE XII: Inference Blending Calibration and Distillation Dilemma Profiling (ResNet9, $\alpha = 0.05$).

Blending / Distillation Scheme	Top-1 Acc	Bottom-10%	ECE (%)	Brier Score
A. Inference Blending & Calibration Modalities				
Logit Blending (Default Proposed)	72.18%	41.90%	13.43%	0.5619
Probability Blending ($T = 1.0$)	72.63%	41.80%	11.81%	0.5954
Temperature Scaled ($T = 0.5$, Sharpened)	72.70%	42.10%	10.84%	0.6213
Temperature Scaled ($T = 1.5$, Smoothed)	72.57%	41.60%	15.61%	0.5901
Temperature Scaled ($T = 2.0$, Diffused)	72.51%	41.20%	19.29%	0.5951
B. The Distillation Dilemma: Output Entropy Profiling				
Unconstrained Local Head Entropy	$H(p_{local}) = 0.732$ nats			
Distilled Local Head Entropy	$H(p_{local}) = 1.084$ nats (+48.0% Entropy Dilution)			

head $w_{l,i}$ and global Root head w_r .

- Intra-Cluster Semantic Sample Pooling under Extreme Skew: Under extreme skew ($\alpha = 0.05$), an isolated edge client possesses only ~ 333 samples per class, leading to severe local overfitting. Update-similarity clustering groups 5–6 clients sharing overlapping classes into the Parent head w_p , aggregating an **intra-cluster sample pool of $6 \times 333 \approx 2000$ distinct samples per class** ($6 \times$ data support) across diverse edge environments. This semantic sample pooling provides robust intermediate representations that mitigate local overfitting while avoiding global FedAvg gradient conflict.

I. Logit Calibration, ECE Analysis, and the Distillation Dilemma

Table XII provides an in-depth characterization of logit calibration across blending modalities and clarifies the theoretical reason why Asymmetric Knowledge Distillation degrades extreme Non-IID performance:

- Calibration across Blending Modalities: Applying Temperature Scaling ($T = 0.5$) during probability blending reduces Expected Calibration Error (ECE) from 13.43% to 10.84% while improving top-1 accuracy to 72.70%. Logit blending achieves the lowest Brier score (0.5619), demonstrating that linear logit combination provides smooth, well-conditioned decision boundaries without requiring separate per-client calibration validation sets.
- Resolving the Distillation Dilemma: In Table IX, adding Root $\rightarrow$ Local asymmetric distillation was shown to reduce Non-IID accuracy (-1.73 pp). This occurs because under extreme statistical skew ($\alpha = 0.05$), an edge client observes only 2–3 active classes. The Root head w_r , trained on global consensus, outputs smooth, high-entropy probability distributions across all $C = 10$ classes. Enforcing $\text{KL}(p_{local} \parallel p_{root})$ penalizes the sharp confidence needed for local decision hyperplanes, increasing local prediction entropy from 0.732 to 1.084 nats (+48.0% entropy dilution). This acts as an artificial regularizing drag on minority classes, proving that unconstrained head optimization is theoretically superior for edge personalization.

V. Discussion

A. The Personalization Trilemma and Multi-Paradigm Synthesis

Our empirical and theoretical findings reveal that effective edge PFL cannot be formulated as optimizing a single isolated metric. Rather, federated deployment is governed by the Personalization Trilemma—the trade-off between statistical accuracy across heterogeneity regimes, on-device memory and compute constraints, and worst-case client fairness. As summarized in Table I, existing paradigms optimize at most two vertices: dual-model regularization (Ditto, APFL) achieves accuracy and fairness at double hardware costs; split-head methods (FedRep, FedPer, FedBABU) optimize memory but suffer IID collapse; clustered FL (CFL) lacks local personalization; and global consensus (FedAvg) collapses under skew. HEP synthesizes the complementary strengths of these four paradigms into a parameter-free, dynamically adaptive framework, providing a balanced operating profile across accuracy, resource efficiency, and tail fairness.

B. Privacy-Preserving Cluster Routing and Gradient Defense

A critical requirement in edge federated deployments is that topology formation must operate without centralizing raw data, feature embeddings, or label distributions. HEP reconciles cluster formation with strict federated privacy through a two-channel architecture:

- **Gradient Inversion Defense via Random Projection:** Projecting classification head updates $\Delta_i^{head} \in \mathbb{R}^{d_{head}}$ ($d_{head} = 2560$) into an $m = 64$ dimensional sketch ($s_i = P\Delta_i^{head}$) compresses the information exposed to the server to 64 linear scalar constraints. Recovering a 3072-dimensional input image from 64 observations is severely underdetermined, preventing gradient matching attacks from reconstructing private client inputs while preserving pairwise cosine clustering geometry.
- **Two-Channel Federated Communication:** The $m = 64$ sketch s_i is transmitted for server-side cluster routing, while full model parameters ($\theta_i, w_{r,i}$) are synchronized via standard Secure Aggregation (SecAg) protocols [11]. This ensures the server learns only the global consensus aggregate without inspecting individual unmasked client weight updates.
- **Differential Privacy and Noise Resilience:** Adding Gaussian perturbation $\mathcal{N}(0, \sigma^2 I)$ to the clipped $m = 64$ sketch yields client-level single-release budgets of $\epsilon_r = 96.9$ and $\epsilon_r = 48.5$ at $\sigma = 0.05$ and $\sigma = 0.10$. Even when strong noise ($\sigma = 1.00$, $\epsilon_r \approx 4.84$) collapses routing stability to 20%, downstream accuracy is protected by the adaptive ensemble routing mechanism, which dynamically concentrates weight on the private Local head ($\alpha_l \approx 0.95$) and anchored Root head, autonomously isolating noisy cluster assignments at a cost of only -0.57 pp.

C. Shared-Backbone Threat Model Scope vs. Dual-Model Edge Costs

Our empirical findings in Table VII elucidate a fundamental architectural boundary:

- **In-Scope Semantic Label Attacks ($\mathcal{T}_{label}$):** Under label-flipping attacks ($f = 40\%$), HEP achieves 70.24% accuracy (+53.59 pp over FedAvg, +7.75 pp over Ditto) without external filters, because head isolation completely shields private classification hyperplanes from poisoned global labels.
- **Out-of-Scope Gradient Poisoning ($\mathcal{T}_{gradient}$):** Under sign-flipping attacks, Ditto achieves higher fault tolerance because it maintains an unshared private model graph. However, this isolation imposes a heavy price: $2\times$ memory (217.15 MB vs 113.42 MB), $2.46\times$ FLOPs, and out-of-memory crashes on memory-constrained edge devices. HEP delivers Pareto-optimal compute/memory efficiency for standard edge deployments where physical nodes are authenticated.
- **Server-Side Filtering Is Not a Remedy:** We empirically rule out the standard counter-approach of replacing FedAvg with Byzantine-robust update statistics (trimmed means, directional trust weighting): under $f = 40\%$ sign-flipping none outperformed naive averaging (Section IV-E1), because tail-splitting defeats rank filtering and honest-heterogeneity centroid decay defeats directional trust. Resilience on shared backbones must therefore come from architectural isolation properties.

D. Class Cardinality, Partial Participation, and Scalability Bounds

Our scalability analysis highlights the operational boundaries of multi-head federated architectures:

- **High Class Cardinality ($C = 100$):** When class cardinality reaches $C = 100$ under extreme skew ($\alpha = 0.05$), edge clients observe narrow class subsets (≤ 6 classes). Integrating Active-Class Logit Masking (ACLM), the entropy skew metric $R_{skew,i}$, and Two-Stage intra-round decoupling eliminates unobserved class negative gradients, allowing HEP to achieve 65.06% personalized accuracy (+3.57 pp over Ditto, +37.39 pp over FedAvg) and 50.17% worst-decile fairness (+2.80 pp over Ditto).
- **Sparse Client Participation ($C_p = 0.20$):** By introducing staleness-aware routing and variance-adaptive cluster momentum ($\beta_{c,k}$), HEP resolves the cluster lag problem on infrequently sampled devices, stabilizing bottom-decile client accuracy under severe skew.

E. Architectural Scope and Foundation Model (LoRA) Extensions

While HEP is concretely evaluated on edge computer vision benchmarks (ResNet-9 and MobileNetV3), its three core architectural pillars are domain- and task-agnostic:

(i) 3-Tier Shared-Backbone Multiplexing, (ii) Binomial Loss Scheduling, and (iii) Adaptive Ensemble Routing. In large language model (LLM) deployments where full dual-model replication (e.g., duplicating an 8B-parameter Transformer) is completely infeasible on consumer edge devices, HEP maps directly to Hierarchical Parameter-Efficient Fine-Tuning (PEFT): edge clients maintain a single frozen shared Transformer backbone paired with three lightweight LoRA adapter modules (Root LoRA for global consensus, Parent LoRA for domain/cluster specialization, and Local LoRA for private edge personalization), with R_{skew} evaluated via domain perplexity divergence.

VI. Conclusion

We presented HEP (Hierarchical Ensemble Personalization), a parameter-efficient, topology-aware hierarchical ensemble framework for personalized federated learning. HEP eliminates manual heuristic tuning by deriving the label skew metric (R_{skew} via normalized entropy), anchor floor (a_i), cluster momentum ($\beta_{c,k}$ via update variance), staleness scale ($\tau_{0,i}$), and logit temperature scaling ($T_{k,i}$) dynamically. Across five heterogeneity regimes on CIFAR-10, HEP demonstrates a Pareto-dominant operating profile across the Personalization Trilemma: it achieves 89.03% personalization accuracy under extreme skew (+0.70 pp over Ditto, +1.96pp over FedRep) at roughly half the memory footprint of dual-model Ditto (113.42 MB vs. 217.15 MB), while completely closing the split-head IID collapse (72.92% vs. FedRep’s 61.79%) and boosting tail-client fairness (+7.17 pp over FedRep under severe skew). On high-cardinality CIFAR-100 ($C = 100$), HEP reaches 65.06% personalized accuracy (+3.57 pp over Ditto) and 50.17% bottom-10% fairness. We coupled random projection sketching for privacy-preserving routing with Secure Aggregation, derived client-level differential privacy accounting, and demonstrated that adaptive ensemble blending provides natural noise resilience. We release the full harness with operator self-verification, cross-backend consistency checks, per-run manifests, and programmatic significance testing to support reproducible edge-federated research.

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Appendix

All experiments in the revised manuscript were executed through a single deterministic harness (scripts/run_comparison.py) with the following guarantees:

A. Device Verification and Cross-Backend Consistency

The evaluation workstation pairs an AMD Radeon GPU with a CPU-only PyTorch build; acceleration is provided by a DirectX-backed backend whose operator coverage is incomplete (e.g., dimension-reduced median returns empty tensors, boolean advanced indexing is unsupported). We therefore (a) restricted all aggregation code to universally supported operators (sort-based statistics, integer `index_select/index_copy`), and (b) added a startup self-check that probes every accelerator-dependent operation used by the training and aggregation paths before accepting the device, falling back to CPU otherwise. An environment variable forces CPU execution for reference runs.

On an identical protocol (seed 42, CIFAR-10, ResNet-9, 15 clients, 10 rounds, $\alpha = 0.5$), accelerated and CPU executions reached 76.07% vs. 76.40% final personalized accuracy (0.33 pp delta; last-5 average delta 0.32 pp), confirming that hardware acceleration does not materially alter reported results.

B. Evaluation-Interval Neutrality

Metrics are evaluated every `eval_interval` rounds plus the final round. Evaluation consumes no random numbers and mutates no model state; skipped rounds log rows without accuracy keys. Consequently this knob changes monitoring granularity only — training trajectories are bit-identical regardless of its value.

C. Hyperparameter Provenance

Every non-standard constant in the released configuration carries a provenance annotation identifying the experiment that validated it. Key entries: binomial anchor $a = 0.15$ (schedule sweep); eval temperatures $(T_l, T_p, T_r) = (0.6, 0.8, 1.0)$ (inference-only calibration study); cluster re-assignment triggers $\bar{S} < 0.30$, misalignment fraction 0.15 (stability study); local compute budget $E = 5$ epochs/client/round — validated to match or exceed $E = 10$ across IID and moderate-skew regimes at half wall-clock cost, keeping HEP’s per-round client compute within $1.67\times$ of split-head baselines training 3 epochs.

D. Statistical Testing

Headline comparisons use paired two-tailed t -tests across seeds 42, 123, 7 at significance level $\alpha = 0.05$ ($df = 2$). Per-seed paired accuracies are retained by the harness, and the significance table is generated programmatically from raw artifacts rather than transcribed.

E. Baseline Fidelity and Provenance

Every competing method was audited against its canonical publication and re-validated inside the released harness; Table III rows are rendered programmatically from raw run artifacts:

- Ditto [4]: the proximal term of the personalization objective anchors at the received global model $w^{(t)}$, matching the original formulation $\min_{w_p} \mathcal{L}(w_p) + \frac{\lambda}{2} \|w_p - w^{(t)}\|_2^2$. The penalty coefficient is fixed at $\lambda = 0.1$ for all clients and rounds (the per-client λ_i selection heuristic of the original work is not reproduced), and both optimization legs share identical local hyperparameters. Historical byzantine-robustness configurations used $\lambda = 0.05$; all re-executed runs standardize on $\lambda = 0.1$.
- APFL [5]: independent dual models with decoupled global tracking and blended-gradient local updates per Algorithm 1 of the original paper; the mixing weight adapts by gradient descent with clamping to $[0.01, 0.99]$, and evaluation blends predictions with the learned weight.
- FedRep [6]: head-first/joint schedule with representation-only aggregation implemented via head-coordinate masking. Two engineering guards deviate deliberately from the canonical description: BatchNorm statistics are paused during head-only phases, and head-phase gradients are clipped (max-norm 1.0) to prevent cold-start classifier saturation

over randomly initialized representations. FedRep receives exactly `local_steps` local passes per round, matching the compute budget of all other baselines.

- FedBABU [8]: canonical protocol — the classification head stays frozen at each client’s initial random initialization during all federated rounds (body-only gradients and uploads) and is trained exclusively at deployment time (`babu_finetune_steps` SGD steps on local training data before evaluation).
- FedPer [7]: joint training with a private final classification layer; all base layers aggregated.
- FedALA [9]: faithful port of the official Adaptive Local Aggregation module — element-wise interpolation weights between the client’s previous local model and the received global model are learned by gradient descent on sampled local data over the top two parameter tensors (sampling rate 80%, $\eta = 1.0$), followed by standard local training and full-model upload.
- CFL [10]: server-side recursive separation driven by pairwise cosine similarities of client updates, with iterative refinement against cluster-mean update directions; the separation threshold ϵ starts high and is successively decreased until the first separation is observed, per the original tuning recipe.

F. Negative-Result Reporting

Section IV-E1 reports server-side Byzantine filtering mechanisms that failed to outperform naive averaging. All tested modes ship as configuration flags in the released code so that reviewers can reproduce the scoreboard directly.